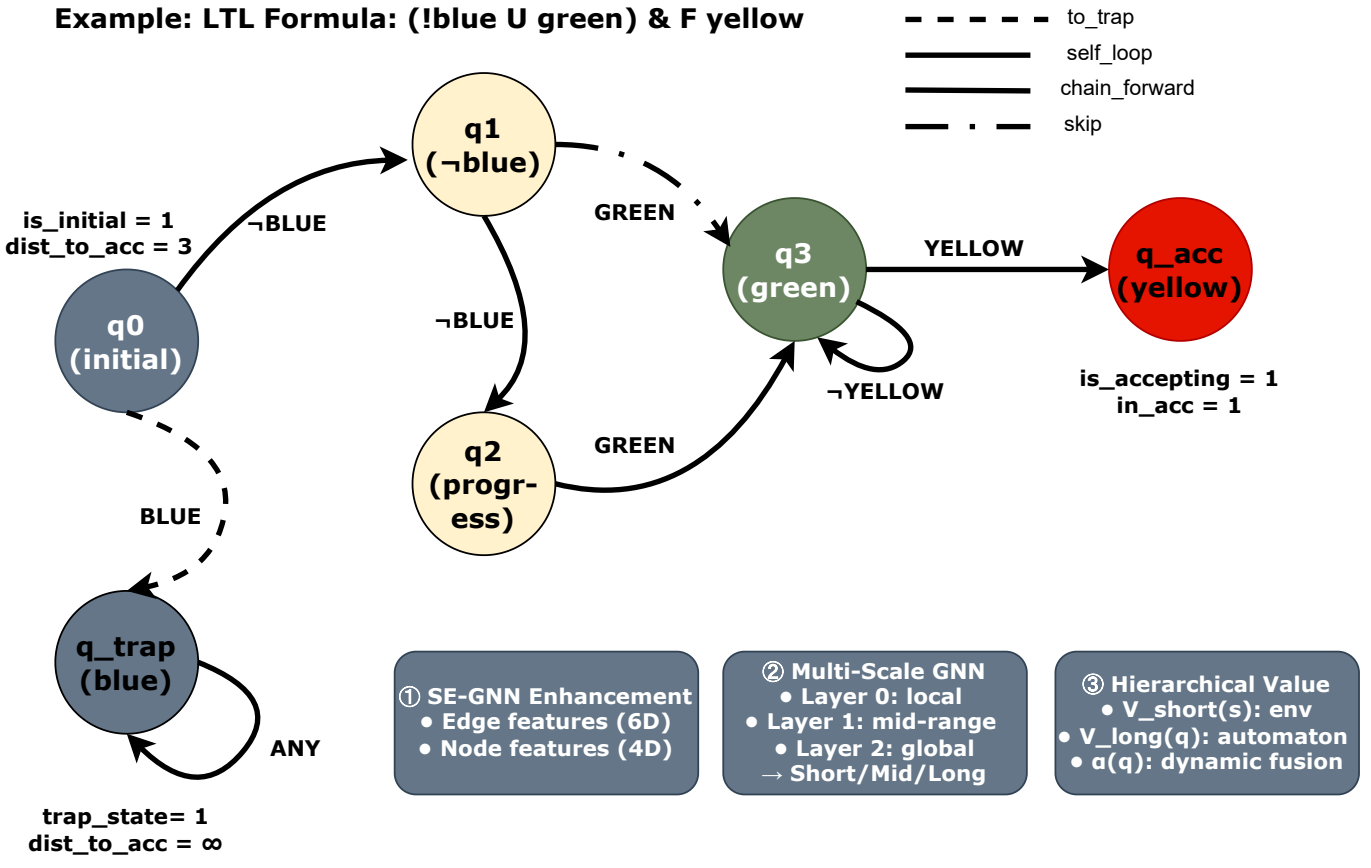

Example: LTL Formula: (!blue U green) & F yellow
to_trap
self_loop
chain_forward
skip
q1
(¬blue)
is_initial = 1
dist_to_acc = 3
¬BLUE
GREEN
q3
(green)
YELLOW
q_acc
(yellow)
q0
(initial)
¬BLUE
¬YELLOW
is_accepting = 1
in_acc = 1
q2
(progr-
ess)
GREEN
BLUE
q_trap
(blue)
ANY
trap_state= 1
dist_to_acc = ∞
① SE-GNN Enhancement
• Edge features (6D)
• Node features (4D)
② Multi-Scale GNN
• Layer 0: local
• Layer 1: mid-range
• Layer 2: global
→ Short/Mid/Long
③ Hierarchical Value
• V_short(s): env
• V_long(q): automaton
• α(q): dynamic fusion